Abdul Rahman Mohamed Farharth

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Research Interests: Biomedical Sensing | Embedded Healthcare Systems | Biomedical Signal Processing | AI for Healthcare

Education

University of Moratuwa, Sri Lanka Jun 2024 – Present
B.Sc. Engineering (Hons.) in Biomedical Engineering,
Department of Electronic and Telecommunication Engineering Expected Graduation: 2028

- CGPA: 3.83/4.00
- Relevant Coursework: Biomedical Instrumentation, Biomedical Device Design, Digital Signal Processing, Image Processing and Machine Vision

Zahira College, Colombo, Sri Lanka 2020 – 2023
G.C.E. Advanced Level Examination – Physical Science Stream

- 3 A passes in Combined Mathematics, Physics, and Chemistry; Island Rank: 407; Z-score: 2.3330

Projects

HCI and Wearable Systems

Smart Posture Monitoring Cushion (P.A.P.A.Y.A.) *Jan 2025 – Apr 2025*

- Developed a real-time posture-monitoring cushion using eight FSR402 pressure sensors and ultrasonic backrest sensing, with vibrotactile feedback for slouch alerts; achieved approximately 70% slouch-detection accuracy.
- Designed the SOLIDWORKS enclosure and sensor-integrated cushion layout and contributed to hardware prototyping, sensor-placement testing, circuit testing, and system integration.
- Conducted a 95-person needs survey on posture-related discomfort to inform the problem definition and design; awarded 2nd Runner-Up at Brainstorm 2025.

Medical AI and Biomedical Signal Processing

EEG-Based Motor Imagery Classification *Aug 2026 – Present*

- Developing an offline motor-imagery BCI pipeline using the PhysioNet EEG Motor Movement/Imagery dataset with MNE-Python, 8–30 Hz filtering, epoch extraction, CSP features, LDA, and stratified cross-validation.
- Evaluating baseline and regularized CSP–LDA configurations and inspecting sensorimotor EEG patterns while extending the project toward more robust subject-level evaluation.

Thyroid Nodule Detection and Malignancy Classification *Jan 2026 – Apr 2026*

- Developed and evaluated a PyTorch-based ultrasound image pipeline for thyroid nodule detection and benign/malignant classification, with an interactive Gradio interface.

Heart Disease Prediction *2026*

- Built a classical machine-learning pipeline on the UCI Cleveland heart-disease dataset with preprocessing, model comparison, 5-fold cross-validation, threshold analysis, and reusable inference; logistic regression achieved 86.9% test accuracy.

Automatic Speaker Recognition System *2025*

- Implemented FFT-based audio preprocessing in MATLAB and integrated the output with an MFCC feature-extraction and speaker-classification pipeline.

Computer Vision and Intelligent Systems

Glove Rejection and Inspection Process (GRIP) *Jan 2026 – Jul 2026*

- Developed a real-time glove inspection and classification system integrating YOLO/OpenCV inference, camera sensing, STM32-based control, embedded communication, and SCARA robotic actuation.
- Contributed to dataset preparation, annotation, model training and validation, real-time inference, telemetry, robustness testing, and hardware-software integration.

Large-Scale License Plate Detection with YOLO11 *2026*

- Fine-tuned YOLO11s on 27,900 images for license-plate localization, reaching mAP50 of 0.918 and precision of 0.956, and extended the pipeline with EasyOCR-based text extraction.

Hand Gesture Recognition with OpenCV and MediaPipe *2026*

- Built a modular real-time hand-tracking and finger-counting system using Python, OpenCV, MediaPipe, and geometric landmark logic.

IoT and Connected Monitoring Systems

Temperature-Controlled Vaccine Cold-Chain Monitoring System — *Jun 2025 – Sep 2025*

- Co-developed a portable vaccine-monitoring system using ESP8266/ESP32-S3, DS18B20 temperature sensing, ESP-NOW wireless communication, configurable thresholds, and audio-visual alerts.
- Contributed to wireless communication architecture, enclosure design, calibration, debugging, and system testing; validated stable transmission to approximately 30 m line-of-sight.

IoT Smart Energy Meter with MQTT and Alerts — *Jul 2025*

- Independently developed an ESP32-based energy-monitoring system using PZEM-004T sensing, embedded C++ firmware, MQTT communication, local display, and automated Telegram/SMS threshold alerts.

Instrumentation, Electronics and Embedded Hardware

Class-D Audio Amplifier — *Jan 2026 – Jun 2026*

- Designed and tested a Class-D amplifier with tone control, approximately 350 kHz PWM generation, MOSFET gate driving, LC output filtering, circuit simulation, and PCB-level implementation.

Analog Automatic Solar Tracker — *Jun 2025 – Nov 2025*

- Co-developed a microcontroller-free single-axis solar tracker using LDR differential sensing, op-amp-based analog PID control, LTspice simulation, PCB implementation, and a SOLIDWORKS mechanical enclosure.

UART Transceiver Implementation in FPGA — *2025*

- Designed and implemented a UART transceiver using Verilog HDL on a DE0 Nano FPGA, focusing on digital communication and timing logic.

Honors and Awards

- **2nd Runner-Up, Brainstorm 2025** – University of Moratuwa engineering innovation competition
- **Dean's List** – Semesters 1, 2 & 4, University of Moratuwa
- **Principal's Gold Medal** – Best Performance in the G.C.E. A/L Physical Science Stream, Zahira College
- **Sri Lanka Mathematics Olympiad Competition 2018** – Distinction Award

Technical Skills

- **Programming:** Python, C, C++, MATLAB
- **Machine Learning & Signal Processing:** PyTorch, scikit-learn, OpenCV, MNE, PyRiemann
- **Embedded Systems:** ESP32, ESP8266, STM32, Arduino, Raspberry Pi
- **Communication:** MQTT, ESP-NOW, I2C, UART
- **Engineering Tools:** Altium Designer, SOLIDWORKS, LTspice, Git

Certificates

- **Machine Learning Specialization** – DeepLearning.AI / Coursera — 2025
- **Medical Image Processing (Biomedical Imaging & Analysis)** – MathWorks / Coursera — 2025

Leadership and Volunteering

- **Finance Lead, Brainstorm 2026; Social Media Manager** – IEEE EMBS Student Branch Chapter — 2025 – Present
- **Event Chair, 5-minute Video Clip** – IEEE Signal Processing Society Student Branch Chapter — 2026 – Present
- **IR Pillar Member; Shuttle Fest 2026 Chairperson** – Electronic Club, University of Moratuwa — 2025 – Present
- **Department Representative** – Department of Electronic and Telecommunication Engineering — 2025

References

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